The Orchestrator: A Lifecycle Architecture for Self-Developing, Self-Operating, and Self-Improving Agentic Systems*

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Abstract

Agentic systems that perform delegated work also need a controlled way to change the concepts and capabilities on which that work depends. This paper proposes a reference architecture organized around DEVELOP, OPERATE, and IMPROVE, coordinated by a durable orchestration layer. Its central unit of deployment is an immutable operating bundle binding executable capabilities to a versioned semantic snapshot, authority policy, evaluation suite, and state compatibility contract. Operational observations challenge the approved model through discrepancy reports, candidate ontology branches, capability development, independent validation, and explicit promotion. The architecture specifies lifecycle transitions, typed handoff contracts, semantic impact obligations, and deployment fencing. A worked example shows how an ontology change can broaden an action's eligibility without changing policy text, while an independent action gate blocks the prohibited effect. Drawing on autonomic computing, runtime models, ontology evolution, and governed capability deployment, the paper connects semantic release identity to authority review and distributed action enforcement. A conditional invariant argument characterizes release integrity, action coherence, and proposal isolation under stated trust assumptions. The contribution is an architectural synthesis supported by illustrative schemas and a falsifiable evaluation design; operational effectiveness and the additional value of ontology-based controls remain unevaluated.

Keywords: agentic systems, ontology evolution, self-adaptive systems, semantic contracts, delegated intelligence, lifecycle governance

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1. Introduction

An agent that can complete a task is not necessarily a system that can maintain the capability to complete that task over time. Procedures become inadequate, observations challenge existing categories, tools change, and the interpretation of a policy may shift when its vocabulary changes. Continuous operation therefore creates a second problem: how to evolve the system without making its operating assumptions untraceable or allowing its effective authority to expand silently.

Consider an automated maintenance system authorized to diagnose faults and perform approved resets on individual machines. It receives a report that two machines repeatedly fail together. Its model treats machines independently, and its diagnostic procedures cannot explain their joint behavior. Investigation reveals a shared controller, requiring the system to extend its ontology with that dependency and develop a coordinated diagnostic capability. Using that capability also requires an explicitly approved scope covering both machines and the controller. The system must preserve the evidence and suspend affected resets while these changes are developed, evaluated, and approved. Learning that the machines share a controller must not, by itself, grant permission to reset that controller or interrupt the other machine.

The research question is: **Can autonomous software systems be organized into distinct development, operation, and improvement functions, coordinated through a shared, versioned ontology, to support continuous evolution while preserving semantic consistency, configuration reproducibility, and controlled authority?** This paper gives a reference architecture and a qualified answer: such a separation can specify enforceable release and action boundaries, provided that the semantic dependencies are actually pinned and the boundaries are mediated by trusted services. It cannot ensure that the model describes the world correctly or that an agent recognizes every unfamiliar situation.

We call the three functions DEVELOP, OPERATE, and IMPROVE. OPERATE performs delegated work using approved capabilities, monitors execution and outcomes, and flags situations that may require intervention by IMPROVE or DEVELOP. IMPROVE investigates operational deficiencies and situations that challenge the current ontology, evaluates new concepts and alternative ontologies, and proposes governed changes to the system's model, capabilities, scope, and permissions. DEVELOP implements and tests executable capabilities against an identified semantic snapshot. An

orchestrator coordinates their transitions and invokes independent release controls. These are responsibilities, not a requirement to run exactly three language models.

In this usage, orchestration includes governance of the semantic models, evidential claims, and task scope through which agents interpret their mandates.

The paper makes four contributions as an architectural synthesis:

1. A deployment contract binding semantics, capabilities, authority, evaluation, and state compatibility into an immutable operating bundle.
2. A lifecycle in which evidence can challenge production semantics through an explicit report, candidate branch, implementation, validation, and promotion sequence.
3. A semantic impact obligation that includes changes in policy denotation and capability applicability, even when policy and capability source files are unchanged.
4. A distributed deployment contract connecting approval to an exact bundle, an activation epoch, compatible state, and action-time enforcement.

The contribution lies in specifying how established mechanisms interact for delegated agentic work: what each transition binds, which authority approves it, and which obligations remain at the point of action. The following sections relate this contract to its architectural foundations and define how its practical value could be tested.

2. Positioning and research method

2.1. Prior architectural foundations

Autonomic computing already treats systems as entities that monitor and adapt their own operation. MAPE-K separates monitoring, analysis, planning, execution, and shared knowledge. DEVELOP/OPERATE/IMPROVE does not replace that control model: OPERATE belongs largely to the managed application; IMPROVE groups analysis and planning; DEVELOP supplies artifacts used in adaptation; orchestration coordinates execution and release. The mapping is many-to-many because monitoring and knowledge management cut across the functions. Kephart and Chess (2003).

IBM's architectural blueprint also discusses orchestrated managers, policy-based delegation, composed management functions, and shared

knowledge with common syntax and semantics. These provide the foundation for treating coordination and semantic interpretation as shared management responsibilities. IBM (2005).

Architecture-based adaptation and models maintained during execution are also established. Rainbow uses an external architecture-based adaptation framework, while models@run.time addresses causally connected models that guide executing systems. They establish the use of an explicit runtime model and a separate adaptation layer. Garlan et al. (2004); Blair, Bencomo, and France (2009).

Ontology evolution research has long examined changes, consistency, dependent artifacts, and propagation. Git-backed RDF version control is likewise established. These sources motivate treating an ontology branch as an engineering artifact whose dependencies and propagation effects require review. Stojanovic et al. (2002); Arndt et al. (2019).

The connection between semantic evolution and security is also older than agentic software. Peldszus and colleagues describe semantic change detection, co-evolution proposals, and security compliance analysis. Our emphasis on authority affected by meaning changes is therefore an application and integration of an existing concern. Peldszus et al. (2021).

2.2. Closest contemporary comparisons

Three recent systems provide particularly close comparisons. Declarative Lifecycle Management for Self-Adaptive Systems connects semantic lifecycle descriptions to reconfiguration of managing-system components. EvoOntology maintains a versioned schema/content/tool ontology layer, uses operational trajectories to propose localized changes, and rejects candidates that fail paired evaluation. Governed Capability Evolution specifies compatibility checks and staged admission, monitoring, withdrawal, and audit for versioned AI components. Kamburjan et al. (2026); Chong et al. (2026); Qin et al. (2026).

Further close comparisons include ontology-aware scientific tool acquisition, a bounded-autonomy workflow framework connecting semantic, evidential, procedural, and authority contracts, and versioned semantic evaluation records. Tang et al. (2026); Jørgensen and Ma (2026); Bousetouane (2026). The last two were screened through primary abstract or indexed publisher content, so detailed absence claims are not justified.

Table 1: Scope of the reference contract in relation to close contemporary systems.

Comparator	Direct overlap	Focus of this reference contract
Declarative Lifecycle Management	Semantic stages and architectural adaptation	Complete release and action binding during model changes
EvoOntology	Versioned schema/content/tools and evidence-based evolution	Import closure, authority consequences, state and activation contracts
Governed Capability Evolution	Policy/behavior/recovery compatibility and staged deployment	Semantic dependencies as part of the same governed candidate
Bounded-Autonomy Framework	Ontology, context, process, and delegated authority contracts	Explicit ontology branch and distributed promotion obligations
EIO-Agents	Versioned evidence meaning and reconstructable evaluation	Operational action records connected to release identity

Table 1 identifies shared mechanisms and the focus of the present specification. Its final column describes design scope; it does not establish that a comparator lacks those features. This paper combines semantic dependency closure, authority consequences, and distributed action admission in a common release contract. The targeted search does not establish priority for that composition.

GitOps supplies a complementary release discipline: declarative desired state, immutable version history, automated reconciliation, and continuous convergence. Software supply-chain provenance contributes artifact lineage and build attestations. Adding an ontology hash to those practices is insufficient when imports, inference configuration, or policy interpretation remain mutable. OpenGitOps principles; SLSA provenance.

2.3. Design method and literature search

The design proceeds in five steps: decompose the lifecycle into responsibilities and trust boundaries; compare those responsibilities with prior architectures; define release identities and transition contracts; challenge the contracts with semantic-change and stale-worker counterexamples; and derive conditional invariants with falsifiable evaluation criteria. The resulting artifacts are a reference architecture, illustrative interface schemas, and an

evaluation design. Literature comparisons establish the design context; the examples and invariant argument provide analytical assessment. They do not substitute for an implementation study.

The supplementary search log records 43 primary discovery queries and nine additional queries submitted on 9 October 2026, sources inspected, access limitations, and a claim-by-claim assessment. Searches covered autonomic computing, ontology evolution, runtime models, knowledge-graph versioning, self-evolving agents, lifecycle governance, and semantic security impact. Close matches were sought deliberately. Foundational studies and recent agent preprints were compared by architectural feature rather than by terminology alone. A focused follow-up on 10 October 2026 examined identity and access control, business-process modeling, AI risk governance, excessive agency, and task alignment; its queries and source-access record appear separately in the search log.

This is a targeted adversarial literature study, not a systematic review with a registered protocol, comprehensive database coverage, or an exhaustive patent search. Search indexing, inaccessible full texts, rapidly revised preprints, and differences between terminology and implementation limit coverage. A feature not observed in an inspected description is recorded as unestablished, not absent. The novelty assessment concludes that the broad thesis is anticipated and that originality of the integrated reference contract remains unestablished. The paper can be evaluated as a design contribution without converting that uncertainty into a claim of discovery.

3. Semantic foundations and problem boundaries

3.1. Distinct artifacts

An **ontology** describes concepts, entity kinds, relations, and applicable axioms. It can be a graph or logical theory; it need not be a tree. A **knowledge base** holds assertions about entities under that vocabulary. **Operational state** records workflow progress, pending actions, leases, and deployed versions. **Observations** are incoming measurements or reports, potentially incomplete or false. **Evidence** preserves sources, retrieval context, and support for claims. **Policies** determine permitted actions and approval conditions. **Capabilities** are executable procedures or services.

These distinctions concern responsibility and meaning, not physical storage. An RDF store may hold both domain definitions and case facts in

different named graphs. A policy can refer to ontology terms without being owned by the process that edits those terms. A statement produced by an LLM can be preserved as a hypothesis without being inserted as an approved fact. An ontology can represent concepts of evidence or authority while records of particular evidence or authority grants remain separate artifacts.

Fact admission also requires controls. An agent that cannot edit the ontology may still change effective permissions by asserting a decision-bearing classification or relationship. The knowledge service therefore authenticates writes, preserves supporting evidence, separates provisional from approved assertions, and applies policy-specific freshness and conflict rules. Routine fact updates need not become ontology releases, but they cannot bypass the evidence requirements for consequential decisions.

The maintenance model might define `Machine`, `Location`, and `locatedIn`. The assertion that machine M17 is in production bay A is a case fact. A maintenance report is evidence supporting observations about its behavior. The requirement for authorization covering every affected asset before a reset is policy. The diagnostic routine is a capability. The paused maintenance case is operational state. Collapsing these into one mutable agent memory would make it difficult to distinguish a new observation from an approved change in meaning.

3.2. A semantic snapshot includes its interpretation

Let an identified semantic snapshot be

$$S = (O, I, H, R, M),$$

where O is the ontology root; I is its transitive, pinned import closure; H is a set of validation shapes; R identifies the entailment regime, reasoner implementation, and configuration; and M contains additional rules, mappings, units, and interpretation conventions needed by consumers. The registry resolves this tuple to immutable bytes and a manifest digest h_S. A Git commit of O alone does not identify S if an imported URL resolves differently tomorrow.

A hash identifies bytes, not equivalence of meaning. Two serializations may encode equivalent axioms while producing different byte hashes. Conversely, an unchanged ontology file may yield different answers after a reasoner or mapping change. A reproducible packaging format must fix file

identities, ordering, digest algorithms, and import resolution. RDF dataset canonicalization can normalize serialization differences and blank-node identifiers; it does not establish logical equivalence between different axiom sets. The canonicalization algorithm and configuration must themselves be identified. W3C RDF Dataset Canonicalization.

OWL entailment and data validation answer different questions. Under open-world reasoning, missing assertions do not generally establish falsity. Some restrictions allow the existence of an unnamed value without showing a stored field is populated. SHACL checks graph conformance under specified targets, constraints, and an explicit entailment setup; conformance is not proof of domain truth. The bundle must state whether shapes examine asserted data, an inferred graph, or another declared view. Reasoner timeout, unsupported semantics, or unknown outcomes cannot be converted into a successful check. W3C OWL 2 Direct Semantics; W3C SHACL.

3.3. What counts as an unfamiliar situation?

The question "What happens when the system encounters something that cannot be represented by its current ontology?" has several answers. The first obligation is to distinguish causes, as summarized in Table 2.

Table 2: Discrepancy classes and their initial operational dispositions.

Discrepancy	Example	Initial disposition
Missing fact	The machine's installed firmware version is unknown	Seek evidence within approved permissions
Unknown entity	A previously unseen asset fits `Machine`	Create a provisional case identity under admission rules
Contradictory evidence	Two sensors report incompatible locations	Preserve sources and investigate; do not overwrite the disagreement
Constraint violation	Authorization for an affected asset is missing	Hold or reject the action under the declared validation policy
Procedure gap	Existing categories fit, but no safe workflow exists	Request a capability change without assuming an ontology change

Discrepancy	Example	Initial disposition
Suspected conceptual gap	A shared-control dependency cannot be represented by the independent-machine model	Report a model limitation and investigate a candidate distinction

An unusual value or a failed procedure is not by itself evidence of a missing concept. A suspected conceptual gap is a defeasible diagnosis: bad data, a poor mapping, ambiguous language, or an unavailable procedure can explain the same symptom. OPERATE can detect residual patterns, validation failures, unexplained tool outcomes, and uncertainty, but cannot reliably recognize everything outside its representational vocabulary. Free text and raw evidence therefore remain available alongside typed fields. Reporting channels must allow "unclassified" rather than forcing every challenge into the very categories under challenge.

4. Governance beyond identity, access control, and workflow

4.1. Complementary control objectives

Established security and business-process approaches provide important foundations for agentic systems. Zero trust emphasizes authenticated, explicitly authorized access to resources under contextual policies. Attribute-based access control (ABAC) evaluates subject, resource, operation, and environmental attributes against policy. Business Process Model and Notation (BPMN) specifies process structure, coordination, and execution semantics. These traditions address complementary control objectives; they do not form an exhaustive partition of agentic-system research. Rose et al. (2020); Hu et al. (2014, updated 2019); Object Management Group (2013).

The architectural distinction drawn here concerns the interpretation supplied to these controls. A valid identity and a permitted tool invocation do not, by themselves, establish that the invocation serves the delegated purpose, that its supporting claims are warranted, or that its target belongs within the intended domain. ABAC can incorporate contextual relationships, and BPMN already has execution semantics. The additional obligation is to govern the domain concepts, evidential assumptions, and scope interpretations on which an agent's selection and justification of actions depend, including changes to those inputs over time.

4.2. Semantic, epistemic, and operational scope

Orchestration in this paper includes maintaining an explicit relationship among three boundaries:

- **Semantic and ontological scope:** the approved entity kinds, relationships, constraints, mappings, and inference rules through which the system interprets its environment. These determine, for example, what counts as a machine, a dependency, or a completed maintenance task, and which physical or digital systems those terms denote.
- **Epistemic scope:** the domains and conditions within which claims may support decisions, the admissible evidence sources, and the requirements for accepting or revising those claims. An observation, a model-generated hypothesis, and a verified fact can warrant different uses. Missing evidence or an unfamiliar situation may require deferral rather than an unsupported inference that expands the task.
- **Task and action scope:** the delegated objectives, named systems and targets, permitted operations, and applicable budgets, time limits, and approval conditions. Establishing a relationship between two assets does not itself extend a delegation to both assets.

These boundaries operationalize concerns already present in AI risk governance. NIST's AI Risk Management Framework calls for documenting intended purposes and deployment contexts (MAP 1.1), supported tasks (MAP 2.1), and knowledge limits and human oversight (MAP 2.2). The proposed architecture binds such constraints to identified instructions, semantic snapshots, evidence-admission policies, grants, and evaluation criteria. Epistemic scope here governs which claims may justify decisions; it neither establishes the truth of every admitted claim nor directly constrains every internal computation of a language model. NIST (2023).

4.3. Scope expansion as a security concern

An agent may widen its interpretation of a task without acquiring a new credential. In the maintenance example, evidence that two machines share a controller may lead the agent to reinterpret diagnosis of one machine as permission to intervene in the coupled system. If available tools have broader privileges than the delegation, or policy decisions depend on classifications the agent can change without independent review, that expansion

can produce unauthorized disclosure, modification, or disruption. The failure concerns the relationship between the delegated purpose and effective action scope; it does not imply that an agent can bypass a correctly enforced resource restriction. OWASP's excessive-agency guidance identifies excessive functionality, permissions, and autonomy as contributing conditions and recommends constrained tools and downstream authorization. OWASP (2025).

Existing agent-security research also addresses this problem beyond identity checks. Task Shield evaluates whether instructions and tool calls contribute to user-defined goals, while ontology-driven security evolution examines security consequences of semantic change. These are direct antecedents, not omissions claimed for prior work. Jia et al. (2024); Peldszus et al. (2021).

The Orchestrator's proposed contribution is to connect these concerns across development, operation, and improvement: changes to meaning, evidence requirements, capabilities, or operating scope become inspectable candidates whose authority consequences require independent assessment. Approved extensions are bound to a release, while consequential actions remain subject to resource-level enforcement. This governance complements authentication, identity management, access control, and workflow coordination. Its benefit is a design hypothesis requiring comparative evaluation; an ontology or an orchestration layer alone is not a security guarantee.

5. Lifecycle responsibilities and the operating bundle

5.1. Three functions and one coordination responsibility

OPERATE accepts work, interprets observations under an approved S, invokes capabilities, and records outcomes. It monitors execution and outcomes, preserves relevant evidence, and flags anomalies or capability gaps that may require IMPROVE's assessment or authorized development by DEVELOP through the orchestrator. It may update permitted case facts and operational state. It cannot overwrite the authoritative semantic snapshot or obtain a release approval merely by declaring that a new interpretation is useful. Its response to uncertainty is configurable: continue unaffected work, invoke a bounded fallback, pause the affected case, or escalate.

IMPROVE assesses operational performance, unfamiliar situations, and candidate extensions or alternatives to the current ontology. It checks competing explanations, gathers permitted evidence, and proposes ontology, mapping, policy, capability, or evaluation changes. It can create a candidate branch and describe a provisional interpretation. Its credentials give it

access to a proposal space, not the production reference. It may recommend no change or reject a misleading report.

DEVELOP builds and tests capabilities against a specific candidate snapshot. Its work order identifies h_S, expected interfaces, permitted development tools, budget, fixtures, and release criteria. If the candidate changes, its outputs and validation evidence must be reconsidered. A tool or prompt update is still a capability change even if no conventional source file changes.

The orchestrator maintains durable lifecycle state and delegates work. Independent services enforce promotion and operational authorization. The orchestrator can be a workflow engine plus supervisors, registries, and deployment controllers. Its reasoning assistant may recommend a transition, but the transition guard executes in a trusted service. Combining functions in one implementation is permitted only if the required credential, approval, and evidence boundaries remain enforced.

5.2. Immutable release identity

Define an operating bundle

$$B = (S, C, T, L, U, P, E, D, X),$$

where C identifies capability code and dependencies; T tool interfaces and implementations; L the model identity and declared runtime characteristics; U prompts and instructions; P authority policy; E evaluations, fixtures, and thresholds; D deployment configuration; and X the operational-state schema and migration/compatibility plan. An immutable manifest gives the bundle digest h_B.

A hosted model alias is not an immutable model identity. If the provider cannot pin a version or supply weights and runtime details, the manifest records that limitation and the known provider revision rather than claiming a fully reconstructible bundle. Tool implementations and remote APIs can also change behind stable identifiers. The release contract must identify these mutable dependencies and the conditions under which their changes trigger review.

For consequential actions, the decision record includes h_B, the activation epoch, the task and observation references, the knowledge-base revision or query snapshot, retrieved semantic slices, policy decision, capability invocation, outputs, and external result. It records what is available, including omissions or redactions. Evidence access is governed separately so that provenance does not become a route to unrestricted disclosure.

This enables **configuration reproducibility** and, where input and state are preserved, **decision reconstruction**. It does not imply identical LLM outputs, repeated external effects, or deterministic replay. Identical configuration can coexist with stochastic decoding, provider changes, nondeterministic tools, and changing external state. A preserved action trace is evidence of an execution, not a promise that rerunning it will repeat the result.

6. Reference architecture and transition contract

6.1. Components and information paths

The reference architecture comprises a durable orchestration service; semantic and capability registries; a version-controlled ontology repository; a shared knowledge service; operational, improvement, and development runtimes; evaluation and policy services; an approval gateway; a deployment controller; an action gateway; and an audit/evidence service. Registries identify approved immutable artifacts. The shared knowledge service serves facts and queries under a specified semantic snapshot and data revision. Observation ingestion stores incoming evidence before interpretation.

Figure 1 shows responsibilities and enforcement paths. The architecture companion provides six additional diagrams detailing lifecycle handoffs, ontology branching, release identity, validation, and governance.

The controller requires durable, idempotent transitions and recoverable scheduling. An event stream can distribute reports and status, but does not by itself provide a unique production reference or enforce approval. A workflow engine can retain lifecycle history, but external actions still need deduplication and reconciliation. Agent supervisors handle bounded tasks and timeouts. Combining these components is a deployment choice; their contract is the design requirement.

6.2. Practical implementation choices

A small implementation could use Git-retained RDF/OWL files with import locks, an independently stored bundle registry, Apache Jena for RDF querying and declared validation/reasoning, and a durable workflow service. RDF supplies the graph representation; OWL supplies selected entailment semantics; SHACL supplies specified graph constraints. They are complementary roles, not interchangeable products. Jena's documented components include Fuseki, persistent stores, inference, and SHACL processing. Apache Jena documentation.

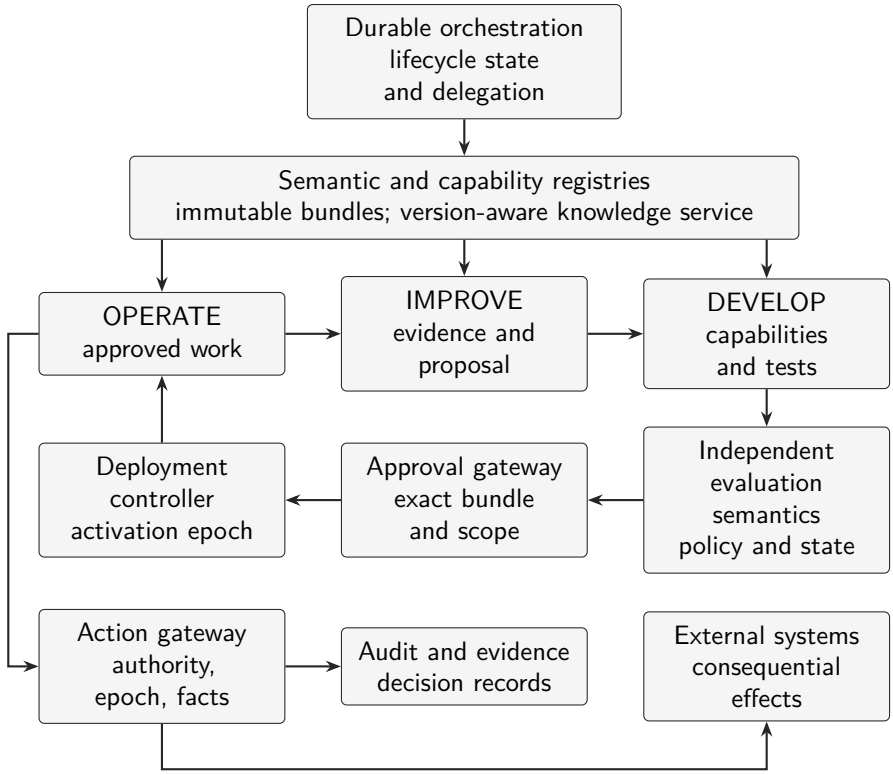


Figure 1: Reference components. Candidate generation and production admission are separate paths. All consequential operational effects pass through the action gateway.

GraphDB and Stardog are alternative knowledge services with documented query and inference facilities. Their repository configuration, inference view, transaction behavior, and artifact export need to be assessed against the bundle contract; a product name does not establish immutable semantic snapshots or promotion independence. WebProtégé is a collaborative authoring option, while a Git-backed knowledge graph such as Quit Store addresses versioned collaboration. Neither an editor's revision history nor a database backup alone identifies the complete operational interpretation. GraphDB documentation; Stardog documentation; WebProtégé project; Arndt et al. (2019).

A durable workflow engine such as Temporal can coordinate long-running activities and retained execution history; an event-driven implementation is also possible if it explicitly supplies equivalent transition durability and reconciliation. The reference architecture requires deterministic transition guards around nondeterministic tasks. Workflow replay does not make an LLM deterministic, and deployment/action authority remains a separate enforcement obligation. These are implementation candidates, not a performance comparison or verified integration. Temporal workflow execution documentation.

6.3. State machine

Let a change request q have a durable revision, parent bundle, candidate digest, evidence references, and lifecycle state. The supplied event schema defines `REPORTED`, `INVESTIGATING`, `PROPOSED`, `BUILD_AUTHORIZED`, `BUILT`, `VALIDATING`, `VALIDATED`, `APPROVED`, `STAGED`, `ACTIVE`, and `RETIRED`, with `REJECTED`, `WITHDRAWN`, `SUPERSEDED`, and `ROLLBACK_PENDING` for alternative paths. A no-change investigation closes as `WITHDRAWN`; an abandoned candidate revision becomes `SUPERSEDED`. Quarantine is an isolation disposition recorded with rejection. Withdrawing an active capability requires recovery or suspension and does not erase its deployment history.

Figure 2 groups related activities into a lifecycle overview. Monitoring occurs within `ACTIVE`; recovery enters `ROLLBACK_PENDING` and must select an approved compatible target. Build and validation each include the finer states defined above. The same path can support routine automated changes under previously approved rules.

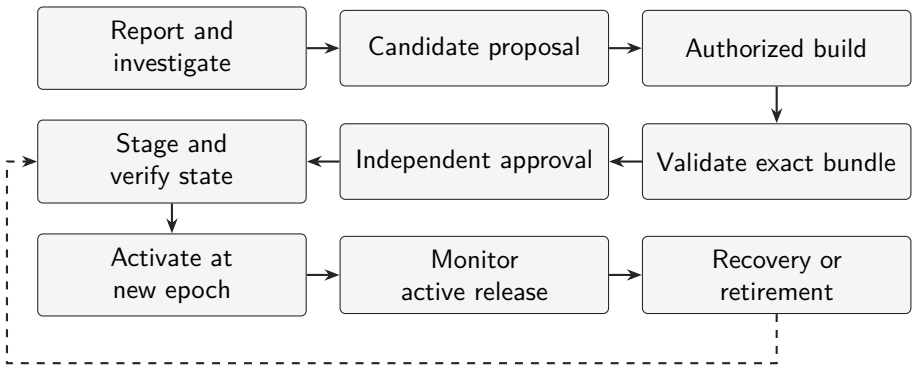


Figure 2: Lifecycle overview; boxes group the finer event-schema states. Each controlled transition retains a guard decision. Development authorization does not authorize deployment.

Table 3: Representative transition guards and required records.

Transition	Required guard and record
Report to investigation	Admitted report, accessible evidence, budget and scope
Proposal to development	Fixed parent and candidate snapshot; authorized work order
Built to validated	Artifact integrity; declared syntax, semantics, behavior, authority, and state checks
Validated to approved	Independent approval over exact bundle and evidence digests; approved scope and expiry
Approved to staged	Deployment credentials; compatible state; verified worker artifacts
Staged to active	Compare-and-swap against expected production epoch; valid approval; action gateway fencing
Active to recovery	Trigger and evidence; authorized compatible target or suspension; external-effect reconciliation

Table 3 summarizes the main guards. Any artifact change invalidates approvals that bind its previous digest. A passing evaluation does not itself

create an approval. Two candidate branches can proceed in parallel, but each is checked against its own parent. If production changes while a candidate is being built, it must be rebased or explicitly approved for the resulting version relationship. Semantic conflicts can exist even when a textual merge is clean.

6.4. Handoff contracts

An `OntologyMismatchReport` binds observations and evidence to the operating bundle, semantic snapshot, trusted runtime context, knowledge read view, and state compatibility epoch. It records provisional discrepancy categories, affected concepts or procedures when known, operational consequence, chosen disposition, and any tentative hypothesis. The `unclassified` category permits a report before a modeled diagnosis exists; `constraint_violation` identifies a declared graph or operational requirement that failed. The report does not grant its author authority to amend production.

An `OntologyChangeProposal` identifies the parent snapshot, candidate, report references, typed changes, rationale, alternative explanations, affected dependents, asserted compatibility, authority impact, migration plan, and required tests. A claim such as "backward compatible" is an input to review rather than a trusted Boolean that skips it.

A `DevelopmentWorkOrder` binds the candidate snapshot to expected behavior, tooling, budget, tests, and the precise authority to build. An `OperatingBundleManifest` pins all release artifacts. A `LifecycleApproval` identifies the bundle, evaluation evidence, governance rule or named approver, scope, activation constraints, and expiry. A `DeploymentRecord` captures bundle activation and recovery. A `DecisionRecord` links accepted or rejected operational actions to their bundle, epoch, state, evidence, and authorization result. Typed JSON Schemas are supplied in the repository; implementations still need to enforce cross-field relationships and credentials.

All mutation APIs use request identities, expected revisions, and authenticated principals. Duplicate delivery must not duplicate a proposal, transition, or external effect. Payload schemas do not prove that evidence is true, that a signature is authentic, or that an emitter has the claimed role. Validation and authentication are separate obligations.

7. Ontology evolution and semantic impact

7.1. Branching and validation

IMPROVE creates a branch from an identified parent rather than modifying production in place. The branch contains proposed ontology axioms, shapes, import pins, mappings, and associated change rationale. Labels such as 1.4.2 and 1.5.0 assist communication; content digests identify actual artifacts. Semantic version numbers are declarations interpreted under a project compatibility policy, not proof of safe substitution.

Validation has several layers. Parsing and schema checks establish usable artifact structure. Logical checks address consistency or selected entailment obligations within a supported logic. Shape checks evaluate declared graph views. Behavioral regression tests exercise affected capabilities. Authority checks examine consequences for authorization. Migration checks assess state compatibility. Evaluation of a candidate must use the exact semantic snapshot and interpretation regime that it would deploy.

The dependency graph connects ontology terms and entailments to queries, classifications, mappings, policies, procedures, datasets, and evaluation fixtures. A semantic diff examines changes in derived conclusions and consumer assumptions, not only inserted triples. Adding a subclass can extend a rule's applicability. Changing a property's domain can introduce classifications through inference. Altering a unit mapping can invalidate a threshold without changing a class hierarchy. Removing a constraint may make a shape pass more cases while reducing the strength of its assurance.

A declared dependency graph can be incomplete, especially when prompts or dynamic queries refer to terms indirectly. Dynamic traces help identify used concepts but miss unexercised paths. Unknown dependencies trigger broader evaluation or a restricted rollout rather than an unsupported declaration of no impact. Semantic comparison depends on the selected logic, signature, and notion of equivalence; computational limits differ across description-logic profiles. The architecture therefore records the supported check and treats inconclusive outcomes as insufficient for automatic admission. Botoeva et al. (2017).

7.2. Authority can change while policy text stays fixed

Let Ω_d be a governance-defined set of fully specified invocations in domain d, identifying each operation, target, and parameter set independently of the candidate ontology. Let $\mathcal{E}(B, K, u) \subseteq \Omega_d$ be the invocations eligible under

classification and procedure-routing rules, and let $A(B, K, u) \subseteq \mathcal{E}(B, K, u)$ be those admitted after the complete policy and action-gateway checks for principal u and approved context K. A comparison across changed identifiers requires a pinned mapping into this common space.

Suppose a routing rule selects `reset(x)` when `Machine(x)` holds. For a fixture containing only `CoupledMachine(x)`, adding

$$\text{CoupledMachine} \sqsubseteq \text{Machine}$$

makes `Machine(x)` derivable and can enlarge $\mathcal{E}$ without changing the policy file. If no other gate excludes the new invocation, A can also expand: $P' = P$ does not imply $A(B', K, u) = A(B, K, u)$. If an independent affected-asset authorization rule rejects it, routing eligibility expands while the action remains inadmissible. The worked example below distinguishes these outcomes.

For automatically promoted changes intended to preserve an existing authority envelope $G(u)$, the release obligation is

$$\forall K \in \mathcal{K}_{\text{approved}} : \quad A(B', K, u) \subseteq G(u).$$

Here $\mathcal{K}_{\text{approved}}$ is a declared scope of contexts; $G(u) \subseteq \Omega_d$ is an authority envelope owned by the governance boundary, not defined solely by the candidate ontology. The envelope is an upper bound; contextual conditions may permit fewer actions. Where comparison with the parent is meaningful, the reviewer also considers $A(B', K, u) \setminus A(B, K, u)$. Expanding that difference is not automatically wrong: an authority owner may authorize a legitimate extension. It is wrong to treat the extension as invisible.

Finite fixture comparisons detect examples, not containment for every future knowledge state. A restricted policy/ontology profile may permit static reasoning over symbolically described contexts. Otherwise the gate needs an explicit scope limitation, conservative rejection of unproven expansion, or an authorized decision that accepts identified uncertainty. Gate criteria and governing vocabulary must themselves be protected from edits by the candidate-producing functions. This obligation applies security co-evolution concerns to the joint semantic release contract; it is not a new theory of access control.

7.3. Knowledge migration and rollback

Ontology evolution may require instance-data migration, reclassification, mapping repair, or quarantine of incompatible records. Migration artifacts bind source and target semantic snapshots, supported state schemas, preserved fields, data-loss conditions, and reconciliation procedures. Historical decisions remain interpreted under their historical bundle, even if current knowledge has migrated. A compatibility view can expose older records, but its semantics and limitations must also be pinned.

Rollback is a new controlled deployment of a compatible prior or repaired bundle at a higher activation epoch. It is not restoration of an earlier epoch value. If a migration was lossy or the world has changed, the old bundle may no longer be safe. External effects such as sending a parcel cannot be undone by restoring code. Recovery may require compensation, a forward repair, suspension, or human intervention. An approval includes the expected recovery route and evidence that it is feasible for the permitted scope, not an unconditional promise to reverse time.

8. Distributed operation, governance, and assurance

8.1. Activation and action-time checks

For each governed domain d, the authoritative production reference is

$$Q_d = (h_B, e, z),$$

where e is a monotonically increasing activation epoch and z identifies the compatible state epoch or schema relationship. The controller stages immutable artifacts, checks approvals and state prerequisites, and commits the reference with compare-and-swap against the expected previous reference. Conflicting controllers cannot both replace the same expected epoch successfully. Epoch-conditioned updates and recipient-side rejection of stale sequencers are established distributed-systems mechanisms; this architecture applies them to the complete semantic operating bundle. Burrows (2006).

Workers verify and load a complete bundle. Each consequential invocation carries its domain, h_B, activation epoch, case-state revision, authenticated principal, and idempotency key. The action gateway compares those values to the authoritative admission rule before authorizing an effect. A stale worker with a perfectly valid old bundle is denied after its authority is

revoked. Merely broadcasting "new ontology available" is insufficient because a partitioned worker can continue using cached semantics.

Logical atomicity applies to the admission boundary, not simultaneous replacement of every distributed process. The companion specifies a base protocol in which only the current bundle epoch admits consequential actions. Retained leases are an optional coexistence extension for rolling deployment: the gateway admits a named, independently authorized lease only while its bundle, scope, expiry, state compatibility, and revocation status remain valid. Deactivation removes an epoch from the default reference; full revocation also invalidates its retained leases. Cached possession of a lease is insufficient. Offline leases would require an expressly accepted revocation-delay bound and cannot provide immediate revocation during disconnection.

The gateway must be at, or trusted by, the effect-producing service. A preflight check followed by an uncontrolled delay is vulnerable to revocation between authorization and execution. An effect token should be consumed under the authoritative epoch or a bounded, expressly authorized lease, and side effects must be idempotent or reconcilable. The service also revalidates decision-bearing facts against their admitted revision or an explicitly authorized freshness window when consuming the token. State-schema compatibility alone does not establish that affected-asset authorization remains valid. No claim of exactly-once effects is made for arbitrary external systems. Where final-effect mediation is impossible, that path falls outside the stated guarantee.

8.2. Roles and autonomy tiers

Distinct principals give OPERATE permission to invoke approved operational tools; IMPROVE permission to investigate and write candidate artifacts; DEVELOP permission to build and test in an isolated environment; and the deployment controller permission to activate approved bundles. The approval gateway evaluates an authenticated governing rule or approver decision. A language model's role description is not an access-control mechanism.

Governance can authorize automatic promotion for a well-defined change class. For example, a documentation-only correction may require immutable provenance and no machine-interpreted change. A restricted semantic extension may require mechanically checked invariants, unchanged effective authority within an approved scope, affected-capability regression evidence, and a reversible state route. A new authorization class or unresolved migration consequence requires its designated authority owner. The architecture does

not require a human signature on every candidate; it requires an approval trace grounded in previously delegated authority.

Human and automated gates both can be wrong. Independent credentials reduce the consequences of a candidate-producing agent being compromised, but do not guarantee correct judgment. Shared ownership, correlated models, misleading evidence, poisoned tests, and compromised gate code remain risks. Evaluation fixtures, gate rules, and promotion credentials must have controlled change paths. A candidate must not be allowed to weaken its own passing criteria and count the resulting pass as independent validation.

8.3. Conditional safety argument

Assume: (a) artifacts identified by digest are immutable and retrievable; (b) authenticated role boundaries and approval integrity hold; (c) the production reference is linearizable per governed domain; (d) trusted runtime mediation attests which bundle and state the worker actually uses, rather than trusting self-reported identifiers; (e) every consequential effect is completely mediated at the effect boundary; and (f) that boundary enforces active epochs or expressly authorized leases, compatible state, and the declared evidence freshness/revalidation rule. These are implementation obligations, not experimental findings.

Three invariants follow from the specified guards:

1. **Release integrity:** every admitted bundle has an approval bound to its exact artifacts, evidence, and scope.
2. **Action coherence:** every admitted action identifies one approved semantic/capability/policy bundle and a compatible state relationship.
3. **Proposal isolation:** candidate writes by IMPROVE or DEVELOP cannot change the authoritative production reference through their ordinary credentials.

A transition argument starts from an approved initial reference. Candidate transitions change only candidate state. Approval stores an exact digest decision. Activation replaces the reference only under a valid guard and expected epoch. An action is admitted only after effect-boundary checks against that reference or an authorized retained lease. Recovery uses the same activation rule at a new epoch. Therefore no specified transition admits an unapproved or internally mixed release, and a revoked epoch cannot authorize a newly mediated action. This immediate-revocation conclusion

excludes offline leases whose expressly accepted revocation delay has not elapsed.

This is a conditional invariant argument over the abstract contract. It is not a machine-checked proof, a guarantee against a faulty gate, or a proof that approved behavior is safe in the world. It excludes unmanaged credentials, unmediated tools, forged state bindings, and uncertain external execution. Semantic consistency is scoped to declared logical and consumer checks. Ontology adequacy, novelty detection, LLM truthfulness, liveness under partition, and universal authority containment are not established by these invariants.

9. Worked example: an unexpected dependency between machines

9.1. Encounter and investigation

A maintenance system runs bundle B_0. Snapshot S_0, labeled 1.4.2, represents machines, locations, fault codes, and maintenance cases. Its diagnostic capability assumes that each case concerns one independently controlled machine. For this constructed scenario, company policy requires authorization covering every affected asset before a reset; unresolved impact requires a hold. The example describes proposed behavior, not an observed deployment.

A maintenance report states that machines M17 and M18 repeatedly fault at the same time. OPERATE preserves the report, emits mismatch report R17 under B_0, and suspends the affected reset workflow. Correlation alone does not establish a shared cause or a missing concept. Other work can continue only where its independence and existing authorization remain established.

IMPROVE investigates with the responsible engineer. Incorrect timestamps, a common power disturbance, missing case facts, or a deficient diagnostic procedure could explain the report. In this case, investigation establishes that M17 and M18 share controller C4 and that the approved model cannot express shared control or its maintenance consequences. If that relationship were already expressible, admitting supported facts could suffice without changing the ontology. Here the proposal adds `Controller`, `controlledBy`, and `CoupledMachine`, and identifies the affected diagnostic queries, reset procedures, and authority checks.

9.2. Candidate development and promotion

A branch from S_0 produces candidate S_1, proposed label 1.5.0. It adds the shared-control relationship, classification rules, and validation shapes for dependency evidence. The statement that M17 and M18 use C4 is admitted as a supported case fact under the new vocabulary. Imports and interpretation settings are pinned. Development is authorized against S_1 while production retains B_0.

DEVELOP builds a coordinated diagnostic capability that correlates records from both machines and their controller, checks the proposed dependency, and produces a maintenance recommendation. This extends the earlier single-machine procedure; adding vocabulary alone does not implement it. The proposed operating scope names M17, M18, and C4. Policy P_1 requires an independently issued grant for the additional diagnostic reads. That grant permits neither controller reset nor interruption of either machine. Any such intervention needs its own authority decision covering the affected assets; the capability cannot grant it to itself.

Tests cover independent machines, coincidental simultaneous faults, confirmed shared control, incomplete dependency evidence, out-of-scope diagnostic access, expired grants, state migration, and stale-worker requests. Expected outcomes precede release. If validation and authority review pass, the independent gateway approves exact bundle B_1. The deployment controller checks state compatibility and advances $Q_{\text{maintenance}}$ from epoch 42 to 43. Action-boundary enforcement rejects obsolete epoch-42 requests. The case resumes with coordinated diagnosis within the granted scope, while resets remain blocked unless separately authorized. The decision record links the anomaly, ontology extension, new capability, scope, and grants.

9.3. A plausible but incompatible candidate

A competing branch adds only `CoupledMachine` as a subclass of `Machine`, retains the single-machine reset procedure, and calls the change additive. Parsing, logical consistency, and generic machine shapes can all pass. Yet the new category inherits reset eligibility under a procedure that assumes an independent target. The affected-asset authorization guard should still block an intervention whose impact or authority is unresolved. That safeguard does not make the candidate procedure compatible.

A differential fixture contains `CoupledMachine(x)` without an assertion of `Machine(x)`. Comparing it before and after the subclass axiom isolates newly inferred reset eligibility. M17 and M18 are separate regression cases:

both were already machines, so the new axiom does not newly classify those same assets. Their cases check that shared control invalidates the independent-reset assumption and requires coordinated diagnosis and an explicit authority review.

The release gate rejects the incomplete candidate. Treating category membership as permission to reset the controller, or checking only the named machine while ignoring affected assets, would violate the stated enforcement assumptions. An unestablished authority obligation prevents automatic promotion. These are analytical checks, not measured detection results; one constructed dependency does not establish coverage of every future anomaly.

10. Failure modes, tradeoffs, and boundaries

Table 4 identifies failures that the release and action contracts must address.

Table 4: Failure modes and the corresponding architectural responses.

Failure mode	Consequence and response
Semantic drift or mutable imports	Same root identifier yields changed conclusions; pin the complete snapshot and invalidate unresolved dependencies
Conflicting branches	Textual merge hides incompatible entailments; re-evaluate merged semantics against current production
Capability incompatibility	Previously handled category now includes untested cases; inspect dependency and assumption impact
Stale workers or version skew	Old semantics authorize actions after revocation; fence at the effect boundary or retain an explicit bounded lease
Invalid migration	New release cannot interpret state; stage against a verified compatibility contract and quarantine unsupported records
Incorrect novelty detection	Ordinary missing data is mistaken for a concept gap; preserve alternatives and permit closure without change
Recursive improvement	Repeated edits consume budget and weaken evaluation; impose work budgets, independent criteria, and stopping rules

Failure mode	Consequence and response
Semantic authority expansion	Classifications broaden permissions without a policy edit; review denotation and protected authority envelopes
Failed rollback	Old bundle cannot recover current state or undo external effects; suspend, compensate, or repair forward
Incomplete ontology	Internal consistency is mistaken for completeness; retain raw evidence and permit challenges outside modeled categories
Model nondeterminism	Reruns differ despite pinned configuration; preserve actual traces and assess outcomes statistically
Communication loss	Reports duplicate or transitions diverge; use durable identities, expected revisions, and reconciliation
Approval bottleneck	Important repairs wait while work is blocked; preauthorize bounded classes and record capacity/latency tradeoffs
Gate or credential compromise	All abstract guarantees can fail; protect the trusted computing base and retain independent audit

A globally coherent ontology is not always desirable or feasible. Different domains may retain different approved snapshots with explicit mappings. A cross-domain handoff binds both bundle identities and the mapping version, and states which authority boundary mediates the resulting action. Domain owners may disagree about categories; an orchestrator cannot resolve institutional disagreement by selecting the highest semantic version. Federation and organizational accountability remain design work.

The architecture incurs storage, validation, and operational latency. Fine-grained manifests and evidence can be expensive; conservative checks can reject useful candidates. Strong fencing may reduce availability during a partition. A shared registry can be a bottleneck or a source of correlated failure. Organizations must decide which actions require the full contract and which bounded exploratory actions may use lighter controls. The consequence of that choice is reduced assurance for the excluded paths, not a universal guarantee at lower cost.

No architecture prevents gradual goal substitution if owners authorize inappropriate changes or all evaluators adopt the same mistaken premise.

Protected budgets, independent criteria, external evidence, and a stop owner reduce opportunities for unchecked recursion. The release contract therefore supplements institutional accountability and cannot replace it.

11. Evaluation agenda and falsifiable expectations

This is a conceptual and architectural paper. The repository contains interface schemas and design documentation, but no complete operational system. The worked example, counterexamples, and invariant argument are analytical artifacts. Performance, detection rates, and operational effectiveness remain open.

A future prototype should implement the registries, durable transitions, candidate isolation, evaluation contracts, migration checks, and mediated action path in a small simulation. A bounded RDF/OWL profile with explicit SHACL and policy semantics would make the assumptions testable before attempting a highly expressive domain. Table 5 defines the main comparisons. Experimental configurations should use the same tasks, models, tools, budgets, and evidence retention where possible.

Table 5: Prospective comparisons and falsifiable expectations; no experimental results are reported.

Comparison	Falsifiable expectation	Measurement
Task-only agent with mutable shared memory	More accepted actions lack reconstructible semantic identity	Missing artifact/state references per consequential action
Versioned ontology plus performance gate	Some semantic authorization expansions survive ordinary quality evaluation	Unsafe activations under seeded hierarchy/mapping changes
Capability governance without complete semantic closure	Mutable imports or inference changes can alter admission under the same root version	Divergent classifications/admission for nominally identical releases
Ordinary schemas, typed APIs, and dependency-aware policy gates	Some ontology-specific benefits may be achievable with cheaper conventional contracts	Matched fault coverage, reconstruction quality, and cost without an ontology
Full reference contract	Targeted faults are rejected or fenced at the stated boundary	Fault coverage, false rejection, migration failures, stale-effect acceptance

These are predictions, not findings. The experiment must include changes outside the test taxonomy, unknown dependencies, incomplete evidence, and policy ambiguities rather than only faults the gate was designed to recognize. Ablations should remove import pinning, authority impact review, state compatibility, and effect-time fencing separately. Otherwise a favorable aggregate result cannot identify the value of the semantic contract.

Quality measures should include task success, proposal acceptance, false discrepancy reports, time to repair, evaluation cost, approval latency, downtime, and recoverability. Security measures should distinguish a candidate's attempted action from an accepted external effect. Reconstruction should be scored by independent reviewers given only retained records. Novelty detection should be evaluated on missing facts, unknown entities, contradictions, and genuine representational gaps, with adjudication of ambiguous cases. Multiple runs and declared uncertainty estimates are needed for stochastic components; repeated tuning on the same holdout should be prevented.

A prospective protocol should freeze the fault taxonomy, adjudication rules, resource budgets, and holdout set before tuning the gates. Paired trials should expose each architecture to the same task, initial facts, and candidate change; repeated model seeds measure variation within a case, rather than supplying independent cases. Include hierarchy extensions, mapping changes, stale evidence, mutable imports, incomplete dependencies, and changes outside the development taxonomy. Independent adjudicators should classify disputed outcomes without knowing which architecture produced them.

Primary measures should distinguish (i) incompatible candidates activated per incompatible candidate submitted, (ii) unauthorized effects accepted per consequential invocation attempted, and (iii) compliant candidates rejected per compliant candidate submitted. Report numerators and denominators, uncertainty across independent cases, and results for each change class. Reconstruction quality, recovery time, evaluation cost, and task completion provide secondary measures. A benefit over the conventional schema-and-policy baseline must remain after matching budgets and enforcement strength; otherwise the results would support release governance without establishing an additional ontology-specific benefit.

The architecture would be practically weakened by evidence that complete bundles are routinely unreconstructible, semantic impact review misses important policy consequences, state compatibility cannot be enforced, gateways permit revoked actions, or validation cost prevents timely adaptation. An adequate evaluation must expose those failure conditions rather than

merely demonstrate successful processing of one return.

12. Relationship to The Orchestrator books

The three books address complementary readers. *Orchestrator: Your Next Job Title* introduces the work of delegated intelligence and the practical obligation to recognize a model limitation, preserve evidence, and route a change. *Delegated Intelligence* describes the contracts and engineering boundaries needed to build and run the system. *Orchestrating Organizations* addresses the ownership, budget, approvals, and stop decisions that give those contracts institutional force. The Orchestrator series.

The ontology concept extends the books' existing distinctions among instructions, memory, authority, evidence, and governance. A shared model gives participants a vocabulary for understanding a case, while an immutable snapshot gives a release a specified semantic context. Neither replaces the delegation contract. The contract still identifies whose work is delegated, which actions are authorized, what evidence returns, and who can stop or change the arrangement.

The books use *orchestrator* for the accountable human professional; this paper uses *orchestration service* for software that executes institutionally authorized transitions. The accompanying integration develops practical discrepancy handling in Book 1, semantic release boundaries in Book 2, and ownership of meaning changes in Book 3. The books motivate the problem, while the technical contract and its limitations are defined independently here.

13. Conclusion

DEVELOP, OPERATE, and IMPROVE provide a useful division of lifecycle responsibility when changes in a system's operating model are treated as governed release events. The shared ontology supplies an explicit semantic reference; the operating bundle binds that reference to the capabilities, policies, evaluations, and state relationships required to use it. A discrepancy can be investigated and modeled without letting a production agent silently redefine the assumptions under which it acts.

The proposed controls establish conditional coherence at the release and action boundary. They do not establish that the ontology is complete, that every novel situation will be detected, or that repeated executions will be

identical. Recent and foundational prior work anticipates much of the design. The research contribution is therefore an integrated reference contract and its exposed obligations, with practical advantage and broader originality left for independent review and empirical evaluation.

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AI assistance. OpenAI Codex assisted with literature discovery, source comparison, architecture drafting, adversarial review, schema preparation, and manuscript editing. The author retains responsibility for the scientific claims, source attribution, and final manuscript.

Artifact and data availability. The manuscript, architecture companion, illustrative interface schemas, fixtures, literature search record, and conceptual mapping to the books are publicly available in the paper repository. The fixtures are constructed examples. No new experimental dataset is reported.

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